

BANK DATABASE: TRIGGERS AND STORED PROCEDURES

Create and Populate Bank Database Tables

The screenshot displays the OneCompiler IDE with a MySQL database. The left pane shows the SQL code for creating and populating the database tables. The right pane shows the results of the queries, including the data inserted into the tables.

SQL Code:

```
CREATE TABLE Customer (
  Customer_ID INT PRIMARY KEY,
  Customer_Name VARCHAR(100) NOT NULL,
  Phone VARCHAR(15),
  Email VARCHAR(100),
  City VARCHAR(50)
);

CREATE TABLE Account (
  Account_No INT PRIMARY KEY,
  Customer_ID INT,
  Account_Type VARCHAR(20),
  Balance DECIMAL(12,2) DEFAULT 0,
  Branch VARCHAR(50),
  FOREIGN KEY (Customer_ID)
    REFERENCES Customer (Customer_ID)
);

CREATE TABLE Bank_Transaction (
  Transaction_ID INT PRIMARY KEY AUTO INCREMENT,
  Account_No INT,
  Transaction_Type VARCHAR(20),
  Amount DECIMAL(12,2),
  Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (Account_No)
    REFERENCES Account (Account_No)
);

CREATE TABLE Loan (
  Loan_ID INT PRIMARY KEY,
  Customer_ID INT,
  Loan_Type VARCHAR(20),
  Loan_Amount DECIMAL(12,2),
  Interest_Rate DECIMAL(5,2),
  FOREIGN KEY (Customer_ID)
    REFERENCES Customer (Customer_ID)
);

INSERT INTO Customer VALUES
(101, 'A. SURESH', '9876543210', 'asuresh@gmail.com', 'Hyderabad'),
(102, 'B. SURESH', '9876543211', 'bsuresh@gmail.com', 'Vijayawada'),
(103, 'C. SURESH', '9876543212', 'csuresh@gmail.com', 'Bangalore'),
(104, 'D. SURESH', '9876543213', 'dsuresh@gmail.com', 'Chennai'),
(105, 'E. SURESH', '9876543214', 'esuresh@gmail.com', 'Hyderabad');

INSERT INTO Account VALUES
(10001, 101, 'Savings', 50000.00, 'Hyderabad'),
(10002, 102, 'Savings', 75000.00, 'Vijayawada'),
(10003, 103, 'Current', 120000.00, 'Bangalore'),
(10004, 104, 'Savings', 45000.00, 'Chennai'),
(10005, 105, 'Current', 30000.00, 'Hyderabad');

INSERT INTO Bank_Transaction VALUES
(1, 10001, 'DEPOSIT', 10000.00, '2020-09-25 17:05:52'),
(2, 10002, 'DEPOSIT', 15000.00, '2020-09-25 17:05:52'),
(3, 10003, 'WITHDRAW', 20000.00, '2020-09-25 17:05:52'),
(4, 10004, 'DEPOSIT', 5000.00, '2020-09-25 17:05:52'),
(5, 10005, 'WITHDRAW', 10000.00, '2020-09-25 17:05:52');

INSERT INTO Loan VALUES
(501, 101, 'Home Loan', 500000.00, 7.50),
(502, 102, 'Education Loan', 100000.00, 6.50),
(503, 103, 'Car Loan', 80000.00, 8.20),
(504, 104, 'Personal Loan', 50000.00, 10.50);
```

Database Results:

Customer Table:

Customer_ID	Customer_Name	Phone	Email	City
101	A. SURESH	9876543210	asuresh@gmail.com	Hyderabad
102	B. SURESH	9876543211	bsuresh@gmail.com	Vijayawada
103	C. SURESH	9876543212	csuresh@gmail.com	Bangalore
104	D. SURESH	9876543213	dsuresh@gmail.com	Chennai
105	E. SURESH	9876543214	esuresh@gmail.com	Hyderabad

Account Table:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	30000.00	Hyderabad

Bank_Transaction Table:

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2020-09-25 17:05:52
2	10002	DEPOSIT	15000.00	2020-09-25 17:05:52
3	10003	WITHDRAW	20000.00	2020-09-25 17:05:52
4	10004	DEPOSIT	5000.00	2020-09-25 17:05:52
5	10005	WITHDRAW	10000.00	2020-09-25 17:05:52

Loan Table:

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	500000.00	7.50
502	102	Education Loan	100000.00	6.50
503	103	Car Loan	80000.00	8.20
504	104	Personal Loan	50000.00	10.50

Q1 — Get All Customers

```
DELIMITER //
```

```
CREATE PROCEDURE GetAllCustomers()
```

```
BEGIN
```

```
    SELECT * FROM Customer;
```

```
END //
```

```
DELIMITER ;
```

```
CALL GetAllCustomers();
```

Customer_ID	Customer_Name	Phone	Email	City
101	K. DEVAJI	9070543108	devaji@gmail.com	Hyderabad
102	K. RITHVICK	9070543101	rithvick@gmail.com	Hyderabad
103	K. SARITHA	9070543102	saritha@gmail.com	Hyderabad
104	K. ARUNITHA	9070543103	arunitha@gmail.com	Hyderabad
105	K. RISHITHA	9070543104	rishitha@gmail.com	Hyderabad

Q2 — Get Account Details

```
DELIMITER //
```

```
CREATE PROCEDURE GetAccountDetails()
```

```
IN p_Account_No INT
```

```
BEGIN
```

```
    SELECT *
```

```
    FROM Account
```

```
    WHERE Account_No = p_Account_No;
```

```
END //
```

```
DELIMITER ;
```

```
CALL GetAccountDetails(10001);
```

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	500000.00	7.50
502	102	Education Loan	100000.00	6.50
503	103	Car Loan	300000.00	8.20
504	104	Personal Loan	500000.00	10.50

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad

Q3 — Get Customer Accounts

```
DELIMITER //
```

```
CREATE PROCEDURE GetCustomerAccounts()
```

```
IN p_Customer_ID INT
```

```
BEGIN
```

```
    SELECT
```

```
        C.Customer_ID,
```

```
        C.Customer_Name,
```

```
        A.Account_No,
```

```
        A.Account_Type,
```

```
        A.Balance,
```

```
        A.Branch
```

```
    FROM Customer C
```

```
    JOIN Account A
```

```
    ON C.Customer_ID = A.Customer_ID
```

```
    WHERE C.Customer_ID = p_Customer_ID;
```

```
END //
```

```
DELIMITER ;
```

```
CALL GetCustomerAccounts(101);
```

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	500000.00	7.50
502	102	Education Loan	100000.00	6.50
503	103	Car Loan	300000.00	8.20
504	104	Personal Loan	500000.00	10.50

Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
101	K. DEVAJI	10001	Savings	50000.00	Hyderabad

Q4 — Deposit Money

```
DELIMITER //
```

```
CREATE PROCEDURE DepositMoney()
```

```
IN p_Account_No INT,
```

```
IN p_Amount DECIMAL(12,2)
```

```
BEGIN
```

```
    UPDATE Account
```

```
    SET Balance = Balance + p_Amount
```

```
    WHERE Account_No = p_Account_No;
```

```
END //
```

```
DELIMITER ;
```

```
CALL DepositMoney(10001,5000);
```

```
SELECT Account_No, Balance
```

```
FROM Account
```

```
WHERE Account_No = 10001;
```

Loan_ID	Customer_ID	Loan_Type
501	101	Home
502	102	Educ
503	103	Car
504	104	Pers

Account_No	Balance
10001	55000.00

Q5 — Withdraw Money

DELIMITER //				3	1000
CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2))				4	1000
BEGIN				5	1000
UPDATE Account					
SET Balance = Balance - p_Amount				Loan_ID	Customer_ID
WHERE Account_No = p_Account_No;				501	101
END //				502	102
DELIMITER ;				503	103
CALL WithdrawMoney(10001,3000);				504	104
SELECT Account_No, Balance					
FROM Account				Account_No	Balance
WHERE Account_No = 10001;				10001	47000.00

Q6 — Prevent Negative Account Balance

DELIMITER //				Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
CREATE TRIGGER CheckBalance				1	10001	DEPOSIT	10000.00	2026-08-25 17:11:05
BEFORE UPDATE ON Account				2	10002	DEPOSIT	15000.00	2026-08-25 17:11:05
FOR EACH ROW				3	10001	WITHDRAW	20000.00	2026-08-25 17:11:05
BEGIN				4	10003	DEPOSIT	5000.00	2026-08-25 17:11:05
IF NEW.Balance < 0 THEN				5	10001	WITHDRAW	10000.00	2026-08-25 17:11:05
SIGNAL SQLSTATE '4000'						Loan_ID	Customer_ID	Loan_Type
SET MESSAGE_TEXT =				501	101	Home Loan	5000000.00	7.50
'Transaction failed: Insufficient balance';				502	102	Education loan	1000000.00	6.50
END IF;				503	103	Car Loan	800000.00	8.20
END //				504	104	Personal loan	500000.00	10.50
DELIMITER ;						10000 1000 (45000) at line 40: Transaction failed: Insufficient balance.		
UPDATE Account								
SET Balance = Balance - 3000								
WHERE Account_No = 10001;								

Q7 — Prevent Invalid Transaction Amount

DELIMITER //				Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
CREATE TRIGGER CheckTransactionAmount				1	10001	DEPOSIT	10000.00	2026-08-25 17:11:05
BEFORE INSERT ON Bank_Transaction				2	10002	DEPOSIT	15000.00	2026-08-25 17:11:05
FOR EACH ROW				3	10001	WITHDRAW	20000.00	2026-08-25 17:11:05
BEGIN				4	10004	DEPOSIT	5000.00	2026-08-25 17:11:05
IF NEW.Amount <= 0 THEN				5	10005	WITHDRAW	10000.00	2026-08-25 17:11:05
SIGNAL SQLSTATE '45000'						Loan_ID	Customer_ID	Loan_Type
SET MESSAGE_TEXT =				501	101	Home Loan	5000000.00	7.50
'Transaction amount must be greater than zero';				502	102	Education loan	1000000.00	6.50
END IF;				503	103	Car Loan	800000.00	8.20
END //				504	104	Personal loan	500000.00	10.50
DELIMITER ;						ERROR 1644 (45000) at line 10: Transaction amount must be greater than zero.		
INSERT INTO Bank_Transaction								
(Account_No, Transaction_Type, Amount)								
VALUES								
(10001, 'DEPOSIT', -3000);								

Q8 — Create Transaction Audit Table

```
CREATE TABLE Transaction_Audit (  
    Audit_ID INT PRIMARY KEY AUTO_INCREMENT,  
    Transaction_ID INT,  
    Account_No INT,  
    Transaction_Type VARCHAR(20),  
    Amount DECIMAL(12,2),  
    Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP  
);  
  
SELECT COUNT(*) AS Total_Audit_Records  
FROM Transaction_Audit;
```

501	101	H
502	102	E
503	103	C
504	104	P

Total_Audit_Records
0

Q9 — Audit New Transactions

```

CREATE TABLE Transaction_Audit (
    Audit_ID INT PRIMARY KEY AUTO_INCREMENT,
    Transaction_ID INT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP
);

SELECT COUNT(*) AS Total_Audit_Records
FROM Transaction_Audit;

DELIMITER //

CREATE TRIGGER AuditTransaction
AFTER INSERT ON Bank_Transaction
FOR EACH ROW
BEGIN
    INSERT INTO Transaction_Audit
    (
        Transaction_ID,
        Account_No,
        Transaction_Type,
        Amount
    )
    VALUES
    (
        NEW.Transaction_ID,
        NEW.Account_No,
        NEW.Transaction_Type,
        NEW.Amount
    );
END //

DELIMITER ;

INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 2000);

SELECT
    Audit_ID,
    Transaction_ID,
    Account_No,
    Transaction_Type,
    Amount
FROM Transaction_Audit;

```

Output:

101	K ANUSRI	9876543210	anusri@gmail.com	Hyderabad
102	K RITHVIK	9876543211	rithvik@gmail.com	Vijayawada
103	M SREEYA	9876543212	sreeya@gmail.com	Bangalore
104	K ADVAITH	9876543213	advaith@gmail.com	Chennai
105	E HEMANTH	9876543214	hemanth@gmail.com	Hyderabad

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-25 17:20:40
2	10002	DEPOSIT	15000.00	2026-08-25 17:20:40
3	10003	WITHDRAW	20000.00	2026-08-25 17:20:40
4	10004	DEPOSIT	5000.00	2026-08-25 17:20:40
5	10005	WITHDRAW	10000.00	2026-08-25 17:20:40

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50

Total_Audit_Records
0

Audit_ID	Transaction_ID	Account_No	Transaction_Type	Amount
1	6	10001	DEPOSIT	2000.00

Q10 — Audit Updated Transactions

```

CREATE TABLE Transaction_Audit (
    Audit_ID INT PRIMARY KEY AUTO_INCREMENT,
    Transaction_ID INT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP
);

SELECT COUNT(*) AS Total_Audit_Records
FROM Transaction_Audit;

DELIMITER //

CREATE TRIGGER AuditTransactionUpdate
AFTER UPDATE ON Bank_Transaction
FOR EACH ROW
BEGIN
    INSERT INTO Transaction_Audit
    (
        Transaction_ID,
        Account_No,
        Transaction_Type,
        Amount
    )
    VALUES
    (
        NEW.Transaction_ID,
        NEW.Account_No,
        NEW.Transaction_Type,
        NEW.Amount
    );
END //

DELIMITER ;

INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 3000);

UPDATE Bank_Transaction
SET Amount = 3500
WHERE Transaction_ID = 6;

SELECT
    Audit_ID,
    Transaction_ID,
    Account_No,
    Transaction_Type,
    Amount
FROM Transaction_Audit;

```

SQL*Plus Output:

```

SQL>
Enter input for the program ( Optional )
>
Output
-----
| 101 | K ANUSRI | 9876543210 | anusri@gmail.com | Hyderabad |
| 102 | K RITHVIK | 9876543211 | rithvik@gmail.com | Vijayawada |
| 103 | M SREEYA | 9876543212 | sreeya@gmail.com | Bangalore |
| 104 | K ADVAITH | 9876543213 | advaith@gmail.com | Chennai |
| 105 | E HEMANTH | 9876543214 | hemanth@gmail.com | Hyderabad |
-----

| Account_No | Customer_ID | Account_Type | Balance | Branch |
|-----|-----|-----|-----|-----|
| 10001 | 101 | Savings | 50000.00 | Hyderabad |
| 10002 | 102 | Savings | 75000.00 | Vijayawada |
| 10003 | 103 | Current | 120000.00 | Bangalore |
| 10004 | 104 | Savings | 45000.00 | Chennai |
| 10005 | 105 | Current | 90000.00 | Hyderabad |
-----

| Transaction_ID | Account_No | Transaction_Type | Amount | Transaction_Date |
|-----|-----|-----|-----|-----|
| 1 | 10001 | DEPOSIT | 10000.00 | 2026-08-25 17:22:35 |
| 2 | 10002 | DEPOSIT | 15000.00 | 2026-08-25 17:22:35 |
| 3 | 10003 | WITHDRAW | 20000.00 | 2026-08-25 17:22:35 |
| 4 | 10004 | DEPOSIT | 5000.00 | 2026-08-25 17:22:35 |
| 5 | 10005 | WITHDRAW | 10000.00 | 2026-08-25 17:22:35 |
-----

| Loan_ID | Customer_ID | Loan_Type | Loan_Amount | Interest_Rate |
|-----|-----|-----|-----|-----|
| 501 | 101 | Home Loan | 5000000.00 | 7.50 |
| 502 | 102 | Education Loan | 1000000.00 | 6.50 |
| 503 | 103 | Car Loan | 800000.00 | 8.20 |
| 504 | 104 | Personal Loan | 500000.00 | 10.50 |
-----

| Total_Audit_Records |
|-----|
| 8 |
-----

| Audit_ID | Transaction_ID | Account_No | Transaction_Type | Amount |
|-----|-----|-----|-----|-----|
| 1 | 6 | 10001 | DEPOSIT | 3500.00 |
|-----|

```

Q11 — Audit Deleted Transactions

<pre> DELIMITER // CREATE TRIGGER AuditTransactionDelete AFTER DELETE ON Bank_Transaction FOR EACH ROW BEGIN INSERT INTO Transaction_Audit (Transaction_ID, Account_No, Transaction_Type, Amount) VALUES (OLD.Transaction_ID, OLD.Account_No, OLD.Transaction_Type, OLD.Amount) END // DELIMITER ; INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 4000); DELETE FROM Bank_Transaction WHERE Transaction_ID = 0; SELECT Audit_ID, Transaction_ID, Account_No, Transaction_Type, Amount FROM Transaction_Audit; </pre>				
Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijaynada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2020-08-25
2	10002	DEPOSIT	15000.00	2020-08-25
3	10003	WITHDRAW	20000.00	2020-08-25
4	10004	DEPOSIT	5000.00	2020-08-25
5	10005	WITHDRAW	10000.00	2020-08-25

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50

Total_Audit_Records
6

Audit_ID	Transaction_ID	Account_No	Transaction_Type	Amount
1	6	10001	DEPOSIT	4000.00

Q12 — Check Account Balance After Transaction

<pre> DELIMITER // CREATE PROCEDURE CheckAccountBalance(IN p_Account_No INT) BEGIN SELECT Account_No, Customer_ID, Account_Type, Balance, Branch FROM Account WHERE Account_No = p_Account_No; END // DELIMITER ; CALL CheckAccountBalance(10001); </pre>				
Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
3	10003	WITHDRAW	20000.00	2020-08-25
4	10004	DEPOSIT	5000.00	2020-08-25
5	10005	WITHDRAW	10000.00	2020-08-25

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad

Q13 — Get Customer Loan Details

<pre> DELIMITER // CREATE PROCEDURE GetCustomerLoanDetails(IN p_Customer_ID INT) BEGIN SELECT C.Customer_ID, C.Customer_Name, L.Loan_ID, L.Loan_Type, L.Loan_Amount, L.Interest_Rate FROM Customer C INNER JOIN Loan L ON C.Customer_ID = L.Customer_ID WHERE C.Customer_ID = p_Customer_ID; END // DELIMITER ; </pre>				
Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2020-08-25 17:26:48
2	10002	DEPOSIT	15000.00	2020-08-25 17:26:48
3	10003	WITHDRAW	20000.00	2020-08-25 17:26:48
4	10004	DEPOSIT	5000.00	2020-08-25 17:26:48
5	10005	WITHDRAW	10000.00	2020-08-25 17:26:48

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50

Customer_ID	Customer_Name	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
101	K.ANUSRI	501	Home Loan	5000000.00	7.50

Q14 — Calculate Total Loan Amount

<pre>SELECT * FROM Bank_Transaction; SELECT * FROM Loan; SELECT SUM(Loan_Amount) AS Total_Loan_Amount FROM Loan;</pre>	<table><tr><th>Total_Loan_Amount</th></tr><tr><td>7300000.00</td></tr></table>	Total_Loan_Amount	7300000.00
Total_Loan_Amount			
7300000.00			

Q15 — Find Highest Loan Amount

70	SELECT * FROM Bank Transaction;	
71	SELECT * FROM Loan;	
72		
73	SELECT	
74	MAX(Loan_Amount) AS Highest_Loan_Amount	5000000.00
75	FROM Loan;	

Q16 — Find Lowest Loan Amount

70	SELECT * FROM Bank_Transaction;	
71	SELECT * FROM Loan;	
72		
73	SELECT	
74	MIN(loan_Amount) AS lowest_loan_Amount	500000.00
75	FROM Loan;	

Q17 — Count Total Customers

<pre>SELECT * FROM Account; SELECT * FROM Bank_Transaction; SELECT * FROM Loan; SELECT COUNT(*) AS Total_Customers FROM Customer;</pre>	<table><tr><th>Total_Customers</th></tr><tr><td>5</td></tr></table>	Total_Customers	5
Total_Customers			
5			

Q18 — Count Customers with Accounts

<pre>SELECT * FROM Bank_Transaction; SELECT * FROM Loan; SELECT COUNT(DISTINCT Customer_ID) AS Customers_With_Accounts</pre>	<table><tr><th>Customers_With_Accounts</th></tr><tr><td>5</td></tr></table>	Customers_With_Accounts	5
Customers_With_Accounts			
5			

Q19 — Find Customers Without Loans

<pre>SELECT C.Customer_ID, C.Customer_Name FROM Customer C LEFT JOIN Loan L ON C.Customer_ID = L.Customer_ID WHERE L.Customer_ID IS NULL;</pre>	<table><tr><th>Customer_ID</th><th>Customer_Name</th></tr><tr><td>185</td><td>E HEMANTH</td></tr></table>	Customer_ID	Customer_Name	185	E HEMANTH
Customer_ID	Customer_Name				
185	E HEMANTH				

Q20 — Find Customers with Savings Accounts

73	SELECT	503	103	Car Loan	100000.00	8.20
74	C.Customer_ID,	504	104	Personal Loan	500000.00	10.50
75	C.Customer_Name,					
76	A.Account_No,			Customer_ID	Customer_Name	Account_No
77	A.Account_Type,			Account_Type	Balance	
78	A.Balance					
79	FROM Customer C			101	K. ARUN	10001
80	INNER JOIN Account A			102	K. KIRAN	10002
81	ON C.Customer_ID = A.Customer_ID			105	E. HEEMATH	10005
82	WHERE A.Account_Type = 'Savings';					

Q21 — Find Customers with Balance Greater Than 50,000

<pre>SELECT C.Customer_ID, C.Customer_Name, A.Account_No, A.Balance FROM Customer C INNER JOIN Account A ON C.Customer_ID = A.Customer_ID WHERE A.Balance > 50000;</pre>	<table><tr><th>Customer_ID</th><th>Customer_Name</th><th>Account_No</th><th>Balance</th></tr><tr><td>102</td><td>K. KIRAN</td><td>10002</td><td>75000.00</td></tr><tr><td>103</td><td>K. SUREYA</td><td>10003</td><td>120000.00</td></tr><tr><td>105</td><td>E. HEEMATH</td><td>10005</td><td>90000.00</td></tr></table>	Customer_ID	Customer_Name	Account_No	Balance	102	K. KIRAN	10002	75000.00	103	K. SUREYA	10003	120000.00	105	E. HEEMATH	10005	90000.00
Customer_ID	Customer_Name	Account_No	Balance														
102	K. KIRAN	10002	75000.00														
103	K. SUREYA	10003	120000.00														
105	E. HEEMATH	10005	90000.00														

Q22 — Find Customers from Hyderabad

<pre>SELECT Customer_ID, Customer_Name, Phone, Email, City FROM Customer WHERE City = 'Hyderabad';</pre>	<table><tr><th>Customer_ID</th><th>Customer_Name</th><th>Phone</th><th>Email</th><th>City</th></tr><tr><td>101</td><td>K. ARUN</td><td>9876543210</td><td>arunr@gmail.com</td><td>Hyderabad</td></tr><tr><td>105</td><td>E. HEEMATH</td><td>9876543214</td><td>heemath@gmail.com</td><td>Hyderabad</td></tr></table>	Customer_ID	Customer_Name	Phone	Email	City	101	K. ARUN	9876543210	arunr@gmail.com	Hyderabad	105	E. HEEMATH	9876543214	heemath@gmail.com	Hyderabad
Customer_ID	Customer_Name	Phone	Email	City												
101	K. ARUN	9876543210	arunr@gmail.com	Hyderabad												
105	E. HEEMATH	9876543214	heemath@gmail.com	Hyderabad												

Q23 — Find Current Account Holders

SELECT	503	103	Car Loan	100000.00	8.20	
C.Customer_ID,	504	104	Personal Loan	500000.00	10.50	
C.Customer_Name,						
A.Account_No,						
A.Account_Type,						
A.Balance						
FROM Customer C		Customer_ID	Customer_Name	Account_No	Account_Type	Balance
INNER JOIN Account A						
ON C.Customer_ID = A.Customer_ID	103	K. SUREYA	10003	Current	120000.00	
WHERE A.Account_Type = 'Current';	105	E. HEEMATH	10005	Current	90000.00	

Q24 — Find Total Balance of All Accounts

<pre>SELECT * FROM Bank_Transaction; SELECT * FROM Loan; SELECT SUM(Balance) AS Total_Account_Balance FROM Account;</pre>	<table><tr><th>Total_Account_Balance</th></tr><tr><td>380000.00</td></tr></table>	Total_Account_Balance	380000.00
Total_Account_Balance			
380000.00			

Q25 — Find Average Account Balance

<pre>SELECT * FROM Bank_Transaction; SELECT * FROM Loan; SELECT AVG(Balance) AS Average_Account_Balance FROM Account;</pre>	<table><tr><th>Average_Account_Balance</th></tr><tr><td>76666.666666</td></tr></table>	Average_Account_Balance	76666.666666
Average_Account_Balance			
76666.666666			

Q26 — Find Number of Loans

<pre>SELECT * FROM Bank_Transaction; SELECT * FROM Loan; SELECT COUNT(*) AS Total_Loans FROM Loan;</pre>	<table><tr><th>Total_Loans</th></tr><tr><td>4</td></tr></table>	Total_Loans	4
Total_Loans			
4			

Q27 — Find Customers with Loans Above 1,000,000

<pre>SELECT C.Customer_ID, C.Customer_Name, L.Loan_ID, L.Loan_Type, L.Loan_Amount FROM Customer C INNER JOIN Loan L ON C.Customer_ID = L.Customer_ID WHERE L.Loan_Amount > 1000000;</pre>	<table><tr><td>502</td><td>102</td><td>Education Loan</td><td>1000000.00</td><td>6.5</td></tr><tr><td>503</td><td>103</td><td>Car Loan</td><td>800000.00</td><td>8.2</td></tr><tr><td>504</td><td>104</td><td>Personal Loan</td><td>500000.00</td><td>10.5</td></tr></table> <table><tr><th>Customer_ID</th><th>Customer_Name</th><th>Loan_ID</th><th>Loan_Type</th><th>Loan_Amount</th></tr><tr><td>101</td><td>K ARUSHI</td><td>501</td><td>Home Loan</td><td>5000000.00</td></tr></table>	502	102	Education Loan	1000000.00	6.5	503	103	Car Loan	800000.00	8.2	504	104	Personal Loan	500000.00	10.5	Customer_ID	Customer_Name	Loan_ID	Loan_Type	Loan_Amount	101	K ARUSHI	501	Home Loan	5000000.00
502	102	Education Loan	1000000.00	6.5																						
503	103	Car Loan	800000.00	8.2																						
504	104	Personal Loan	500000.00	10.5																						
Customer_ID	Customer_Name	Loan_ID	Loan_Type	Loan_Amount																						
101	K ARUSHI	501	Home Loan	5000000.00																						

Q28 — Find Customers with Loans Below 1,000,000

M SELECT					503	103	Car Loan	800000.00	8.20
C.Customer_ID,					504	104	Personal Loan	500000.00	10.50
C.Customer_Name,									
L.Loan_ID,									
L.Loan_Type,									
L.Loan_Amount									
FROM Customer C,					Customer_ID	Customer_Name	Loan_ID	Loan_Type	Loan_Amount
INNER JOIN Loan L					103	M SHEENA	503	Car Loan	800000.00
ON C.Customer_ID = L.Customer_ID					104	K ARUSHI	504	Personal Loan	500000.00
WHERE L.Loan_Amount < 1000000;									

Q29 — Find Loans with Interest Rate Above 7%

<pre>SELECT C.Customer_ID, C.Customer_Name, L.Loan_ID, L.Loan_Type, L.Loan_Amount, L.Interest_Rate FROM Customer C INNER JOIN Loan L ON C.Customer_ID = L.Customer_ID WHERE L.Interest_Rate > 7;</pre>	<table><tr><td>501</td><td>101</td><td>Car Loan</td><td>800000.00</td><td>8.20</td></tr><tr><td>504</td><td>104</td><td>Personal Loan</td><td>500000.00</td><td>10.50</td></tr></table> <table><tr><th>Customer_ID</th><th>Customer_Name</th><th>Loan_ID</th><th>Loan_Type</th><th>Loan_Amount</th><th>Interest_Rate</th></tr><tr><td>101</td><td>K ARUSHI</td><td>501</td><td>Home Loan</td><td>5000000.00</td><td>7.50</td></tr><tr><td>103</td><td>M SHEENA</td><td>503</td><td>Car Loan</td><td>800000.00</td><td>8.20</td></tr><tr><td>104</td><td>K ARUSHI</td><td>504</td><td>Personal Loan</td><td>500000.00</td><td>10.50</td></tr></table>	501	101	Car Loan	800000.00	8.20	504	104	Personal Loan	500000.00	10.50	Customer_ID	Customer_Name	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate	101	K ARUSHI	501	Home Loan	5000000.00	7.50	103	M SHEENA	503	Car Loan	800000.00	8.20	104	K ARUSHI	504	Personal Loan	500000.00	10.50
501	101	Car Loan	800000.00	8.20																															
504	104	Personal Loan	500000.00	10.50																															
Customer_ID	Customer_Name	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate																														
101	K ARUSHI	501	Home Loan	5000000.00	7.50																														
103	M SHEENA	503	Car Loan	800000.00	8.20																														
104	K ARUSHI	504	Personal Loan	500000.00	10.50																														

Q30 — Find Loans with Interest Rate Below 7%

```

SELECT
  L.Customer_ID,
  L.Customer_Name,
  L.Loan_ID,
  L.Loan_Type,
  L.Loan_Amount,
  L.Interest_Rate
FROM Customer C
INNER JOIN Loan L
  ON C.Customer_ID = L.Customer_ID
WHERE L.Interest_Rate < 7;

```

Customer_ID	Customer_Name	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
181	Amit	181	Home Loan	1000000.00	7.50
182	Rishi	182	Education Loan	1000000.00	6.50
183	Sriya	183	Car Loan	1000000.00	6.20
184	Amit	184	Personal Loan	100000.00	10.50

Q31 — Find Customers with More Than One Account

```

CREATE TABLE Loan (
  Customer_ID INT NOT NULL,
  FOREIGN KEY (Customer_ID)
  REFERENCES Customer (Customer_ID)
);

INSERT INTO Customer VALUES
(181, 'Amit', '9876543210', 'amit@gmail.com', 'Hyderabad'),
(182, 'Rishi', '9876543211', 'rishi@gmail.com', 'Vijaywada'),
(183, 'Sriya', '9876543212', 'sriya@gmail.com', 'Bangalore'),
(184, 'Amit', '9876543213', 'amit@gmail.com', 'Chennai'),
(185, 'Rishi', '9876543214', 'rishi@gmail.com', 'Hyderabad');

INSERT INTO Account VALUES
(10001, 181, 'Savings', 50000, 'Hyderabad'),
(10002, 182, 'Savings', 75000, 'Vijaywada'),
(10003, 183, 'Current', 120000, 'Bangalore'),
(10004, 184, 'Savings', 45000, 'Chennai'),
(10005, 185, 'Current', 90000, 'Hyderabad');

INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),
(10002, 'DEPOSIT', 15000),
(10003, 'WITHDRAW', 20000),
(10004, 'DEPOSIT', 5000),
(10005, 'WITHDRAW', 10000);

INSERT INTO Loan VALUES
(181, 181, 'Home Loan', 1000000, 7.5),
(182, 182, 'Education Loan', 1000000, 6.5),
(183, 183, 'Car Loan', 1000000, 6.2),
(184, 184, 'Personal Loan', 100000, 10.5);

SELECT
  COUNT(*) AS Customers_With_Multiple_Accounts
FROM (
  SELECT Customer_ID
  FROM Account
  GROUP BY Customer_ID
  HAVING COUNT(*) > 1
) AS MultipleAccounts;

```

Output

Customers_With_Multiple_Accounts
1